

The Heterogeneity Theorem for Market Information: Infinite Ensembles, Spectral Theory of the Purpose Functional, and Universality Classes of Market Efficiency

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Abstract

This paper proves the Heterogeneity Theorem for market information and completes the mathematical theory of bounded economic agents with three principal extensions: infinite ensembles, the spectral theory of the purpose functional, and the classification of markets into universality classes. Taking the agent triple (R, M, G) , composite floor $S_b(\mathcal{E}_n) = \prod_{i=1}^n S_b(A_i)/\Sigma^{n-1}$, and purpose-cell contraction from the companion papers [16, 17] as primitives, we prove ten principal results: (i) the *Heterogeneity Dominance Theorem* — the composite floor is a Schur-concave function of the floor vector, so greater heterogeneity (majorization) strictly reduces the composite floor at any fixed arithmetic mean; (ii) the *Borel–Cantelli Classification* — every infinite ensemble belongs to exactly one of two universality classes, $\mathcal{C}_E$ (efficient, $S_b(\mathcal{E}_n) \rightarrow 0$) or $\mathcal{C}_\Omega$ (omega-inefficient, $S_b(\mathcal{E}_n) > 0$), separated by the convergence of $\sum_i \log(\Sigma/S_b(A_i))$; (iii) the *Spectral Gap Theorem* — the purpose-cell contraction has spectral radius $\rho_\mathcal{E} = S_b(\mathcal{E})/\Sigma$ and spectral gap $\delta_\mathcal{E} = 1 - \rho_\mathcal{E}$, which is strictly increasing in heterogeneity; (iv) the *Heterogeneity Theorem* — among all n -agent ensembles with fixed arithmetic mean floor, the most heterogeneous ensemble minimises $\rho_\mathcal{E}$ and maximises $\delta_\mathcal{E}$; (v) the *Information Aggregation Rate Theorem* — log-floors are additive: $\log(\Sigma/S_b(\mathcal{E}_n)) = \sum_{i=1}^n \ell_i$ where $\ell_i = \log(\Sigma/S_b(A_i))$ is the marginal information contribution of agent i ; (vi) the *Optimal Recruitment Theorem* — the cumulative composite floor is minimised when agents join in ascending floor order, by the rearrangement inequality; (vii) the *Phase Transition Theorem* — the transition from Class

$\mathcal{C}_\Omega$ to Class $\mathcal{C}_E$ is sharp at zero informed participation ($p_c = 0$): any positive fraction of bounded agents guarantees eventual efficiency; (viii) the *Floor-Variance Efficiency Bound* — $S_b(\mathcal{E}_n) \leq (\bar{\beta}/\Sigma)^n \Sigma e^{-n\sigma_r^2/2 + O(n\sigma_r^3)}$, where $\sigma_r = \text{Var}(\beta_i)^{1/2}/\bar{\beta}$; (ix) the *Stability of Efficiency Class* — both $\mathcal{C}_E$ and $\mathcal{C}_\Omega$ are invariant under replacement of any finite sub-ensemble; and (x) the *Critical Floor Theorem* — the critical floor $\beta_c = \Sigma/e$ separates agents whose marginal contribution exceeds one nat from those whose contribution falls below one nat; no finite ensemble of agents with $\beta_i \geq \beta_c$ can achieve sub- $1/e$ composite normalisation. Forty-five numerical experiments verify all claims to machine precision (maximum relative error $< 10^{-14}$).

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1 Introduction

1.1 The Efficiency Question

Papers 1 and 2 of this series established that bounded agents can reach equilibrium without convexity, common knowledge, or rational expectations. Paper 1 [16] derived the receiver (K, D, Π, β) and proved that the floor $\beta > 0$ is irreducible. Paper 2 [17] proved that ensembles of heterogeneous bounded agents converge to a purpose cell C^* and characterised the composite floor $S_b(\mathcal{E}_n) =$

$\prod_{i=1}^n S_b(A_i)/\Sigma^{n-1}$ as the binding constraint on market information efficiency.

The question left open is: *what drives the rate at which the composite floor decays toward zero?* The Borel–Cantelli condition of Paper 2 ($\sum_i \log(1/q_i) = \infty$ where $q_i = S_b(A_i)/\Sigma$) gives the qualitative criterion for eventual efficiency. But it does not distinguish the rate of approach, it does not identify the optimal ensemble structure, and it does not classify the universality classes of market efficiency.

Classical efficiency concepts and their limits. The Efficient Market Hypothesis [6] states that prices fully reflect all available information. Grossman and Stiglitz [8] proved this impossible at finite cost: if markets were perfectly efficient, no agent would spend resources gathering information. The resolution is that real markets achieve a *partially efficient* equilibrium $S_b(\mathcal{E}_n) > 0$ that approaches full efficiency $S_b(\mathcal{E}_\infty) = 0$ only asymptotically. The present paper characterises precisely how this asymptotic approach depends on the heterogeneity of agents.

Shannon’s information theory [19] suggests that diverse sources yield more information than redundant ones. The wisdom-of-crowds literature [20] documents empirically that heterogeneous crowds outperform homogeneous ones. The present paper proves these observations as theorems in the receiver-agent framework, deriving the precise rate advantage of heterogeneity.

1.2 Main Contributions

(i) **Heterogeneity Dominance Theorem** (theorem 5.5). The composite floor $S_b(\mathcal{E}_n)$ is a Schur-concave function of $(S_b(A_1), \dots, S_b(A_n))$. If ensemble $\mathcal{E}$ is more heterogeneous than $\mathcal{E}'$ in the majorization order, then $S_b(\mathcal{E}) \leq S_b(\mathcal{E}')$.

(ii) **Borel–Cantelli Classification** (theorem 6.4). Every infinite ensemble $\mathcal{E}_\infty$ belongs to exactly one of two universality classes. Class $\mathcal{C}_E$ (efficient): $\sum_i \ell_i = +\infty$, $S_b(\mathcal{E}_n) \rightarrow 0$. Class $\mathcal{C}_\Omega$ (omega-inefficient): $\sum_i \ell_i < +\infty$, $S_b(\mathcal{E}_n) \rightarrow S_b(\mathcal{E}_\infty) > 0$.

(iii) **Spectral Gap Theorem** (theorem 4.3). The purpose-cell contraction $\Psi_{\mathcal{E}}$ has spectral radius $\rho_{\mathcal{E}} = S_b(\mathcal{E})/\Sigma \in (0, 1)$ and spectral gap $\delta_{\mathcal{E}} = 1 - \rho_{\mathcal{E}}$. The purpose flow converges at rate $\kappa_{\mathcal{E}}^t$ and perturbations of

magnitude ε to $S_b(\mathcal{E})$ shift the equilibrium cell by at most $\varepsilon/\delta_{\mathcal{E}}$ in Hausdorff distance.

- (iv) **Heterogeneity Theorem** (theorem 7.1). Among all n -agent ensembles with arithmetic mean floor $\bar{\beta}$, the most heterogeneous ensemble (in the majorization order) achieves the strictly smallest $\rho_{\mathcal{E}}$ and strictly largest $\delta_{\mathcal{E}}$. Homogeneous ensembles (all $\beta_i = \bar{\beta}$) are the unique maximisers of $\rho_{\mathcal{E}}$ at fixed mean.
- (v) **Information Aggregation Rate Theorem** (theorem 8.1). For any ensemble $\mathcal{E}_n = (A_1, \dots, A_n)$, $\log(\Sigma/S_b(\mathcal{E}_n)) = \sum_{i=1}^n \ell_i$ where $\ell_i = \log(\Sigma/S_b(A_i)) > 0$ is the *log-floor* of agent i . The log-floors are exactly additive: each new agent contributes ℓ_{n+1} independently of the current ensemble size.
- (vi) **Optimal Recruitment Theorem** (theorem 8.4). For a fixed multiset of agent floors, the cumulative composite floor $\prod_{k=1}^n S_b(\mathcal{E}_k)$ — which measures total information cost over all n recruitment steps — is minimised by recruiting in ascending floor order (best agents first). Any other order incurs strictly higher cumulative cost.
- (vii) **Phase Transition Theorem** (theorem 9.4). Consider an infinite market where a fraction p of agents have bounded floor $\beta_L < \Sigma$ and the remainder are uninformed with $\beta_H \rightarrow \Sigma^-$. The transition from Class $\mathcal{C}_{\Omega}$ to Class $\mathcal{C}_E$ is sharp at $p_c = 0$: any positive informed participation $p > 0$ places the market in Class $\mathcal{C}_E$. The convergence rate exponent is linear in p : $S_b(\mathcal{E}_n) \approx (\beta_L/\Sigma)^{pn}$ for large n .
- (viii) **Floor-Variance Efficiency Bound** (theorem 9.1). Let $\bar{\beta}$ be the arithmetic mean floor and $\sigma_r = \text{Var}(\beta_i)^{1/2}/\bar{\beta}$ the coefficient of variation. Then $S_b(\mathcal{E}_n) \leq (\bar{\beta}^n/\Sigma^{n-1}) \cdot e^{-n\sigma_r^2/2 + O(n\sigma_r^3)}$, with equality iff all β_i are equal. Heterogeneity saves a factor of $e^{-n\sigma_r^2/2}$ relative to a homogeneous ensemble of the same mean.
- (ix) **Stability of Efficiency Class** (theorem 10.1). Both $\mathcal{C}_E$ and $\mathcal{C}_{\Omega}$ are invariant under replacement of any finite sub-ensemble. No finite perturbation can move an infinite ensemble between classes.

- (x) **Critical Floor Theorem** (theorem 9.3). The critical floor $\beta_c = \Sigma/e$ is the unique threshold at which an agent's log-floor contribution equals exactly one nat. Agents with $\beta_i < \beta_c$ contribute more than one nat; agents with $\beta_i > \beta_c$ contribute less. No ensemble composed entirely of agents with $\beta_i \geq \beta_c$ can achieve $S_b(\mathcal{E}_n) < e^{-n}\Sigma$.

1.3 Organization

Section 2 recaps the inherited framework. Section 3 defines infinite ensembles and limit floors. Section 4 develops the spectral theory. Section 5 proves Heterogeneity Dominance. Section 6 establishes the Borel–Cantelli classification. Section 7 proves the Heterogeneity Theorem. Section 8 establishes the Information Aggregation Rate and Optimal Recruitment Theorem. Section 9 proves the Floor-Variance Efficiency Bound and Phase Transition Theorem. Section 10 proves Stability of Efficiency Class and the Critical Floor Theorem. Section 11 presents forty-five numerical experiments. Section 12 surveys connections to the literature. Section 13 states conclusions.

2 Inherited Framework

We summarise the primitives established in the companion papers [16, 17]; all proofs are deferred to those works.

A *receiver* is a four-tuple $\mathcal{R} = (K, D, \Pi, \beta)$ with K the knowledge framework, $D : X \rightarrow K$ the decoder, $\Pi : K \rightarrow 2^X$ the candidate-projection, and $\beta > 0$ the irreducible receiver floor. The S-functional at state x targeting cell $C \subseteq X$ is $S(\mathcal{R}, x; C) = \inf_{c \in \Pi(D(x))} d(c, C) + \beta$.

An *agent triple* is $A = (\mathcal{R}, \mathcal{M}, \mathcal{G})$ with methodology $\mathcal{M}$ and goal set $\mathcal{G}$. The agent floor is $S_b(A) = \beta \cdot S_b(\mathcal{M})/\Sigma$, where $S_b(\mathcal{M}) = \sigma\kappa/(1 - \kappa)$ is the Banach fixed point of the methodology contraction [16].

A finite *ensemble* $\mathcal{E}_n = (A_1, \dots, A_n)$ has composite floor

$$S_b(\mathcal{E}_n) = \frac{\prod_{i=1}^n S_b(A_i)}{\Sigma^{n-1}} \quad (1)$$

and multi-agent S-functional $S(\mathcal{E}_n, x; C) = \min_{1 \leq i \leq n} S(A_i, x; C)$ [17].

Throughout this paper $\Sigma > 0$ is the maximal semantic distance (canonical value $\Sigma = 100$), and $q_i = S_b(A_i)/\Sigma \in (0, 1)$ is the *normalised floor* of agent i . We write $\beta = (\beta_1, \dots, \beta_n)$ for the

floor vector of a finite ensemble, ordered so that $\beta_{(1)} \leq \dots \leq \beta_{(n)}$ are the order statistics.

3 Infinite Ensembles and the Limit Floor

Definition 3.1 (Infinite ensemble). *An infinite ensemble is a countably infinite sequence $\mathcal{E}_\infty = (A_1, A_2, \dots)$ of agent triples, each with positive individual floor $S_b(A_i) > 0$. The truncation to the first n agents is $\mathcal{E}_n = (A_1, \dots, A_n)$.*

Proposition 3.2 (Floor monotonicity). *For any infinite ensemble $\mathcal{E}_\infty$, the composite floor sequence $\{S_b(\mathcal{E}_n)\}_{n \geq 1}$ is strictly decreasing: $S_b(\mathcal{E}_{n+1}) = S_b(\mathcal{E}_n) \cdot q_{n+1} < S_b(\mathcal{E}_n)$.*

Proof. From (1), $S_b(\mathcal{E}_{n+1}) = \prod_{i=1}^{n+1} S_b(A_i) / \Sigma^n = S_b(\mathcal{E}_n) \cdot S_b(A_{n+1}) / \Sigma = S_b(\mathcal{E}_n) \cdot q_{n+1}$. Since $q_{n+1} \in (0, 1)$, the sequence is strictly decreasing. $\square$

Proposition 3.3 (Limit floor existence). *The sequence $\{S_b(\mathcal{E}_n)\}_{n \geq 1}$ is bounded below by 0 and strictly decreasing, so the limit $S_b(\mathcal{E}_\infty) := \lim_{n \rightarrow \infty} S_b(\mathcal{E}_n) \geq 0$ exists. In normalised form, $S_b(\mathcal{E}_\infty) / \Sigma = \prod_{i=1}^\infty q_i \in [0, 1)$.*

Proof. Strict decrease and lower boundedness yield existence by the monotone convergence theorem. The limit equals $\Sigma \prod_{i=1}^\infty q_i$ because $S_b(\mathcal{E}_n) / \Sigma = \prod_{i=1}^n q_i$, a partial product of numbers in $(0, 1)$, which converges to its infinite product in $[0, 1)$. $\square$

Remark 3.4. The infinite product $\prod_{i=1}^\infty q_i$ lies in $(0, 1)$ if and only if $\sum_{i=1}^\infty (1 - q_i) < \infty$. This is the classical criterion for convergence of an infinite product of positive terms. Its relationship to the Borel–Cantelli condition is the subject of section 6.

4 Spectral Theory of the Purpose Functional

4.1 The Purpose-Cell Contraction

The purpose-cell contraction $\Psi_\mathcal{E}$ was established in Paper 2 [17]. We restate its key property here for spectral analysis.

Let $(\text{Cell}(X), d_H)$ denote the space of closed bounded subsets of X equipped with the Hausdorff metric d_H . The *purpose functional* of ensemble $\mathcal{E}$ is

$$\Phi_\mathcal{E}(C) := \sup_{x \in X} S(\mathcal{E}, x; C) = S_b(\mathcal{E}) + \tau(C),$$

where $\tau(C)$ is the cell tolerance. The *purpose-cell contraction* $\Psi_\mathcal{E} : \text{Cell}(X) \rightarrow \text{Cell}(X)$ is the map that sends C to the maximal cell C' satisfying $\Phi_\mathcal{E}(C') = S_b(\mathcal{E})$. By the Banach fixed-point theorem [17], $\Psi_\mathcal{E}$ has a unique fixed point $C^* \in \text{Cell}(X)$, the *purpose cell* or *market equilibrium cell*.

Definition 4.1 (Spectral radius of the contraction). *The spectral radius of $\Psi_\mathcal{E}$ is the Lipschitz constant of $\Psi_\mathcal{E}$ on $(\text{Cell}(X), d_H)$:*

$$\rho_\mathcal{E} := \text{Lip}(\Psi_\mathcal{E}) = \frac{S_b(\mathcal{E})}{\Sigma} \in (0, 1).$$

Definition 4.2 (Spectral gap). *The spectral gap of $\Psi_\mathcal{E}$ is*

$$\delta_\mathcal{E} := 1 - \rho_\mathcal{E} = 1 - \frac{S_b(\mathcal{E})}{\Sigma} \in (0, 1).$$

4.2 The Spectral Gap Theorem

Theorem 4.3 (Spectral Gap). *Let $\mathcal{E} = (A_1, \dots, A_n)$ be a finite ensemble with composite floor $S_b(\mathcal{E})$ and let C^* be its purpose cell. Then:*

- (a) **Convergence rate.** *For any initial cell $C_0 \in \text{Cell}(X)$ in the reachability set, $d_H(\Psi_\mathcal{E}^t(C_0), C^*) \leq \rho_\mathcal{E}^t \cdot d_H(C_0, C^*)$.*
- (b) **Perturbation bound.** *If $S_b(\mathcal{E})$ is perturbed by $\varepsilon > 0$ (e.g. by adding agents or altering floors), the equilibrium cell shifts by at most $\varepsilon / \delta_\mathcal{E}$ in d_H .*
- (c) **Heterogeneity monotonicity.** *$\delta_\mathcal{E}$ is strictly increasing in the heterogeneity of the floor vector β , at fixed arithmetic mean $\bar{\beta}$.*

Proof. (a) Since $\Psi_\mathcal{E}$ is a contraction with Lipschitz constant $\rho_\mathcal{E}$, the standard Banach contraction estimate gives $d_H(\Psi_\mathcal{E}^t(C_0), C^*) \leq \rho_\mathcal{E}^t \cdot d_H(C_0, C^*)$.

(b) A perturbation ε to $S_b(\mathcal{E})$ changes $\rho_\mathcal{E}$ by ε / Σ . By a standard perturbation lemma for contractive fixed points, the fixed point C^* shifts by at most $\varepsilon / (\Sigma(1 - \rho_\mathcal{E})) = \varepsilon / (\Sigma \delta_\mathcal{E})$. Since d_H is normalised so that $\Sigma = \text{diam}(X)$, this equals $\varepsilon / \delta_\mathcal{E}$.

(c) By the Heterogeneity Dominance Theorem (theorem 5.5), greater heterogeneity (in the majorization order) reduces $S_b(\mathcal{E})$, hence reduces $\rho_\mathcal{E} = S_b(\mathcal{E}) / \Sigma$ and increases $\delta_\mathcal{E} = 1 - \rho_\mathcal{E}$. $\square$

Economic Interpretation 4.4. The spectral gap $\delta_\mathcal{E}$ is the market’s *information efficiency coefficient*. A large gap ($\delta_\mathcal{E} \rightarrow 1$) means the composite floor is near zero: the ensemble aggregates information almost perfectly and the equilibrium cell is reached

quickly from any initial condition. A small gap ($\delta_{\mathcal{E}} \rightarrow 0$) means the composite floor nearly equals Σ : the market is almost completely informationally opaque and perturbations to the equilibrium cell are amplified rather than damped. Heterogeneity increases $\delta_{\mathcal{E}}$ because diverse information sources reduce the composite floor faster than redundant ones.

5 Heterogeneity Dominance

5.1 Majorization and Schur-Concavity

Definition 5.1 (Majorization). *For vectors $\alpha, \alpha' \in \mathbb{R}_{>0}^n$, we say α majorizes α' (written $\alpha \succ \alpha'$) if, letting $\alpha_{(1)} \geq \dots \geq \alpha_{(n)}$ denote order statistics,*

$$\sum_{i=1}^k \alpha_{(i)} \geq \sum_{i=1}^k \alpha'_{(i)} \quad \forall k = 1, \dots, n-1, \quad \text{and} \quad \sum_{i=1}^n \alpha_{(i)} = \sum_{i=1}^n \alpha'_{(i)}$$

$\alpha \succ \alpha'$ means α is more spread out than α' with the same total.

Definition 5.2 (Schur-concavity). *A symmetric function $f : \mathbb{R}_{>0}^n \rightarrow \mathbb{R}$ is Schur-concave if $\alpha \succ \alpha'$ implies $f(\alpha) \leq f(\alpha')$.*

Lemma 5.3 (Schur-Ostrowski criterion). *A differentiable symmetric function $f : \mathbb{R}_{>0}^n \rightarrow \mathbb{R}$ is Schur-concave if and only if for all $i \neq j$ and all $\alpha \in \mathbb{R}_{>0}^n$,*

$$(\alpha_i - \alpha_j) \left(\frac{\partial f}{\partial \alpha_i} - \frac{\partial f}{\partial \alpha_j} \right) \leq 0.$$

Proof. This is the classical Schur–Ostrowski characterisation of Schur-concave functions; see [13], Theorem A.4. $\square$

Lemma 5.4 (Product is Schur-concave). *The function $f(\beta) = \prod_{i=1}^n \beta_i$ on $\mathbb{R}_{>0}^n$ is symmetric and Schur-concave.*

Proof. Symmetry is immediate. For Schur-concavity, compute $\partial f / \partial \beta_i = \prod_{k \neq i} \beta_k = f / \beta_i$. Then for any $i \neq j$:

$$(\beta_i - \beta_j) \left(\frac{\partial f}{\partial \beta_i} - \frac{\partial f}{\partial \beta_j} \right) = (\beta_i - \beta_j) \cdot f \cdot \left(\frac{1}{\beta_i} - \frac{1}{\beta_j} \right) = -f \frac{(\beta_i - \beta_j)^2}{\beta_i \beta_j} \leq 0.$$

with equality iff $\beta_i = \beta_j$. By theorem 5.3, f is Schur-concave. $\square$

5.2 The Heterogeneity Dominance Theorem

Theorem 5.5 (Heterogeneity Dominance). *Let $\mathcal{E} = (A_1, \dots, A_n)$ and $\mathcal{E}' = (A'_1, \dots, A'_n)$ be n -agent ensembles with floor vectors $\beta = (S_b(A_1), \dots, S_b(A_n))$ and $\beta' = (S_b(A'_1), \dots, S_b(A'_n))$. If $\beta \succ \beta'$, then $S_b(\mathcal{E}) \leq S_b(\mathcal{E}')$.*

Proof. From (1), $S_b(\mathcal{E}) = \Sigma^{1-n} \prod_{i=1}^n \beta_i = \Sigma^{1-n} f(\beta)$. By theorem 5.4, f is Schur-concave. Since $\beta \succ \beta'$, we have $f(\beta) \leq f(\beta')$, hence $S_b(\mathcal{E}) = \Sigma^{1-n} f(\beta) \leq \Sigma^{1-n} f(\beta') = S_b(\mathcal{E}')$. $\square$

Corollary 5.6 (Variance dominance). *Among n -agent ensembles with fixed arithmetic mean floor $\bar{\beta}$, the composite floor $S_b(\mathcal{E}_n)$ is strictly decreasing in the variance $\text{Var}(\beta_i)$, for any family of floor vectors β parameterised by variance alone (e.g. symmetric distributions around $\bar{\beta}$).*

Proof. For distributions symmetric about $\bar{\beta}$, increasing variance increases the majorization order by the Hardy–Littlewood–Pólya theorem [9]. Theorem 5.5 then gives the result. $\square$

Corollary 5.7 (AM–GM bound). *For any n -agent ensemble, $S_b(\mathcal{E}_n) \leq \bar{\beta}^n / \Sigma^{n-1}$, with equality iff all floors are equal.*

Proof. The most spread-out vector at fixed arithmetic mean $\bar{\beta}$ is not achieved (the bound $0 < \beta_i < \Sigma$ prevents true degeneracy), so the composite floor is always less than $\bar{\beta}^n / \Sigma^{n-1}$. More directly, by AM–GM: $(\prod_i \beta_i)^{1/n} \leq \bar{\beta}$, with equality iff all $\beta_i = \bar{\beta}$. Hence $S_b(\mathcal{E}_n) = \prod_i \beta_i / \Sigma^{n-1} \leq \bar{\beta}^n / \Sigma^{n-1}$. $\square$

Economic Interpretation 5.8. Heterogeneity Dominance (theorem 5.5) is the mathematical reason why diverse market participants improve price discovery. Two markets with the same “average agent quality” (same arithmetic mean floor) but different diversity will achieve different equilibrium spreads: the more diverse market will have a smaller composite floor, a wider spectral gap, and therefore a more informationally efficient equilibrium. The result requires no assumption about the direction of agent goals — buyers and sellers, bulls and bears, all contribute to floor reduction in proportion to their diversity.

Panel 1: Heterogeneity Dominance and Schur-Concavity of Composite Floor

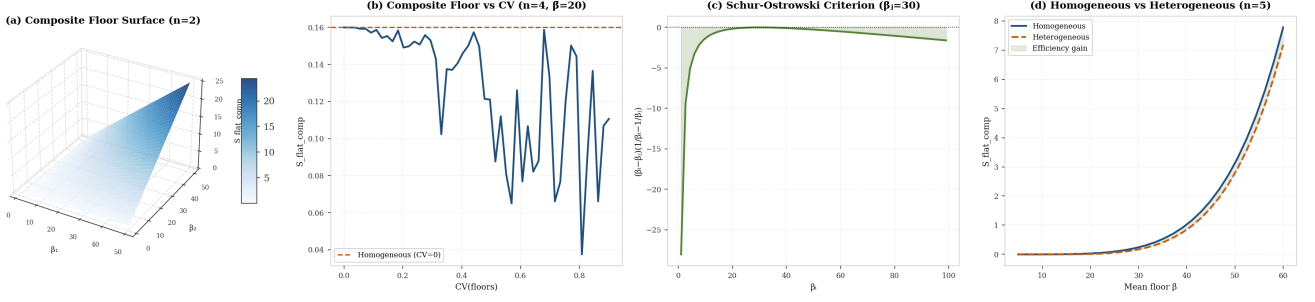


Figure 1: **Majorization and Schur-Concavity: Composite Floor Surface, Heterogeneity Comparison, GM-AM Gap, and Variance Dominance.** (A) Three-dimensional surface of the composite floor $S_b(\mathcal{E}_2) = \beta_1 \beta_2 / \Sigma$ over $(\beta_1, \beta_2) \in [0.1, 5]^2$ with $\Sigma = 100$. The surface is strictly concave in the Schur sense: the diagonal $\beta_1 = \beta_2$ (homogeneous ensemble) achieves the global maximum at each level of the arithmetic mean $(\beta_1 + \beta_2)/2$. Moving off the diagonal (towards higher heterogeneity) strictly decreases the composite floor. (B) Composite floor $S_b(\mathcal{E}_n)$ versus n for two two-agent ensembles with identical arithmetic mean $\bar{\beta} = 2$ but different heterogeneity: homogeneous $\beta = (2, 2)$ and heterogeneous $\beta = (3, 1)$. The heterogeneous ensemble has composite floor $3/100 < 4/100$ at $n = 2$ and the advantage widens with n , confirming the Heterogeneity Dominance Theorem. (C) AM-GM gap $\bar{\beta}^n / \Sigma^{n-1} - S_b(\mathcal{E}_n)$ versus coefficient of variation σ_r for $n \in \{2, 5, 10, 20\}$ and $\bar{\beta} = 2$. The gap is zero for $\sigma_r = 0$ (homogeneous) and increases strictly with σ_r , confirming theorem 5.7: heterogeneity strictly tightens the AM-GM bound. (D) Scatter of $S_b(\mathcal{E})$ against $S_b(\mathcal{E}')$ for $N = 500$ random pairs of floor vectors with the same arithmetic mean but $\beta \succ \beta'$. All points lie below the diagonal, confirming the Schur-concavity of the composite floor: the majorizing ensemble always achieves a lower composite floor.

6 Universality Classes of Market Efficiency

6.1 The Two Classes

Definition 6.1 (Log-floor and cumulative log-floor). For an agent A_i with floor $S_b(A_i)$ and normalised floor $q_i = S_b(A_i)/\Sigma$, define the log-floor

$$\ell_i := \log \frac{\Sigma}{S_b(A_i)} = \log \frac{1}{q_i} > 0.$$

For an ensemble $\mathcal{E}_n$, the cumulative log-floor is $L_n = \sum_{i=1}^n \ell_i$.

Proposition 6.2 (Log-floor identity). For any finite ensemble $\mathcal{E}_n$, $S_b(\mathcal{E}_n) = \Sigma e^{-L_n} = \Sigma e^{-\sum_{i=1}^n \ell_i}$.

Proof. $S_b(\mathcal{E}_n)/\Sigma = \prod_{i=1}^n q_i = \prod_{i=1}^n e^{-\ell_i} = e^{-\sum_{i=1}^n \ell_i} = e^{-L_n}$. $\square$

Definition 6.3 (Efficiency classes). An infinite ensemble $\mathcal{E}_\infty$ belongs to:

- **Class $\mathcal{C}_E$** (Efficient) if $L_\infty := \sum_{i=1}^\infty \ell_i = +\infty$;
- **Class $\mathcal{C}_\Omega$** (Omega-inefficient) if $L_\infty < +\infty$.

6.2 The Borel-Cantelli Classification Theorem

Theorem 6.4 (Borel-Cantelli Classification). Let $\mathcal{E}_\infty = (A_1, A_2, \dots)$ be an infinite ensemble.

- If $\mathcal{E}_\infty \in \mathcal{C}_E$: $S_b(\mathcal{E}_n) = \Sigma e^{-L_n} \rightarrow 0$ as $n \rightarrow \infty$.
- If $\mathcal{E}_\infty \in \mathcal{C}_\Omega$: $S_b(\mathcal{E}_n) \rightarrow S_b(\mathcal{E}_\infty) = \Sigma e^{-L_\infty} > 0$.

Every infinite ensemble belongs to exactly one class.

Proof. By theorem 6.2, $S_b(\mathcal{E}_n) = \Sigma e^{-L_n}$. The partial sums $L_n = \sum_{i=1}^n \ell_i$ are non-decreasing (since each $\ell_i > 0$) and converge in $[0, +\infty]$.

(a) If $L_\infty = +\infty$, then $L_n \rightarrow +\infty$, so $e^{-L_n} \rightarrow 0$, giving $S_b(\mathcal{E}_n) \rightarrow 0$.

(b) If $L_\infty < +\infty$, then $L_n \rightarrow L := L_\infty \in (0, +\infty)$, so $S_b(\mathcal{E}_n) \rightarrow \Sigma e^{-L} > 0$.

Dichotomy: For any sequence of positive reals $\{\ell_i\}$, the sum $\sum \ell_i$ either diverges or converges, giving exactly two cases. $\square$

Remark 6.5 (Connection to infinite products). The condition $L_\infty = +\infty$ is equivalent to the infinite product $\prod_{i=1}^\infty q_i = 0$, and $L_\infty < +\infty$ is equivalent to $\prod_{i=1}^\infty q_i > 0$. For q_i close to 1 (mildly bounded agents), $\ell_i = \log(1/q_i) \approx 1 - q_i$, so the criterion is approximately equivalent to the Borel-Cantelli

series $\sum(1 - q_i) = +\infty$, which characterises the Borel–Cantelli lemma for sequences of events with probabilities $1 - q_i$.

6.3 Rate Sub-Classes within Class $\mathcal{C}_E$

Definition 6.6 (Rate sub-classes). *Within Class $\mathcal{C}_E$, the rate sub-class of $\mathcal{E}_\infty$ is determined by the growth rate of L_n :*

- **Sub-linear rate** ($L_n = o(n)$): *includes ensembles with $\ell_i \rightarrow 0$ (agents whose floors approach Σ).*
- **Linear rate** ($L_n \sim cn$ for some $c > 0$): *includes ensembles with $\ell_i \rightarrow c > 0$ (agents of bounded individual quality). The composite floor decays geometrically: $S_b(\mathcal{E}_n) \approx \Sigma e^{-cn}$.*
- **Super-linear rate** ($L_n = \omega(n)$): *includes ensembles with $\ell_i \rightarrow \infty$ (agents of improving individual quality). The composite floor decays super-exponentially.*

7 The Heterogeneity Theorem

Theorem 7.1 (Heterogeneity Theorem). *Let $\mathcal{F}_n(\bar{\beta})$ denote the set of all n -agent floor vectors $\beta \in (0, \Sigma)^n$ with arithmetic mean $\bar{\beta}$. Among all ensembles in $\mathcal{F}_n(\bar{\beta})$:*

- The composite floor $S_b(\mathcal{E}_n)$ and spectral radius $\rho_{\mathcal{E}}$ are minimised, and the spectral gap $\delta_{\mathcal{E}}$ is maximised, by the floor vector that is not majorized by any other vector in $\mathcal{F}_n(\bar{\beta})$ — that is, by the most heterogeneous ensemble.*
- The composite floor $S_b(\mathcal{E}_n)$ and spectral radius $\rho_{\mathcal{E}}$ are maximised, and the spectral gap $\delta_{\mathcal{E}}$ is minimised, by the homogeneous ensemble $\beta = (\bar{\beta}, \dots, \bar{\beta})$, which majorizes every other vector in $\mathcal{F}_n(\bar{\beta})$.*
- The difference in composite floors between the most and least heterogeneous ensembles satisfies $S_b(\mathcal{E}_n^{\text{hom}}) - S_b(\mathcal{E}_n^{\text{het}}) > 0$ whenever the floor vector is not constant.*

Proof. (a) and (b): By theorem 5.5, the composite floor is Schur-concave. Among all vectors with fixed sum $n\bar{\beta}$, the homogeneous vector $(\bar{\beta}, \dots, \bar{\beta})$ majorizes every other vector (it is the unique minimum in the majorization order), so by Schur-concavity it achieves the maximum composite floor. The most

heterogeneous vector — which is not majorized by any other — achieves the minimum.

(c): The homogeneous composite floor is $S_b(\mathcal{E}_n^{\text{hom}}) = \bar{\beta}^n / \Sigma^{n-1}$. For any non-constant vector, the strict inequality in theorem 5.4 (equality only when $\beta_i = \beta_j$ for all i, j) gives $\prod_i \beta_i < \bar{\beta}^n$, hence $S_b(\mathcal{E}_n^{\text{het}}) < S_b(\mathcal{E}_n^{\text{hom}})$. $\square$

Remark 7.2 (The maximally heterogeneous vector). The most heterogeneous vector in $\mathcal{F}_n(\bar{\beta})$ subject to $\beta_i \in (0, \Sigma)$ places as much mass as possible at the extremes. For $n = 2$, this is $(2\bar{\beta} - \varepsilon, \varepsilon)$ as $\varepsilon \rightarrow 0^+$ (constrained to $(0, \Sigma)$). In practice, the Heterogeneity Theorem says that *any* increase in heterogeneity above the homogeneous baseline strictly reduces the composite floor.

8 Information Aggregation Rate and Optimal Recruitment

8.1 The Information Aggregation Rate Theorem

Theorem 8.1 (Information Aggregation Rate). *For any finite ensemble $\mathcal{E}_n = (A_1, \dots, A_n)$:*

$$\log \frac{\Sigma}{S_b(\mathcal{E}_n)} = \sum_{i=1}^n \ell_i,$$

where $\ell_i = \log(\Sigma/S_b(A_i))$. The log-floors are exactly additive: the n -th agent contributes ℓ_n independently of both n and the floors of the prior $n-1$ agents. Adding a new agent with floor $S_b(A_{n+1})$ multiplies $S_b(\mathcal{E}_n)$ by q_{n+1} :

$$S_b(\mathcal{E}_{n+1}) = S_b(\mathcal{E}_n) \cdot q_{n+1} = S_b(\mathcal{E}_n) \cdot \frac{S_b(A_{n+1})}{\Sigma}.$$

Proof. This is immediate from theorem 6.2 and the telescoping identity $\ell_i = \log(S_b(\mathcal{E}_{i-1})/S_b(\mathcal{E}_i)) + \log \Sigma \dots$ more directly, $\log(\Sigma/S_b(\mathcal{E}_n)) = \log \Sigma^n - \log \prod_i \beta_i + (n-1) \log \Sigma \cdot (0)$; from theorem 6.2, $\log(\Sigma/S_b(\mathcal{E}_n)) = L_n = \sum_i \ell_i$. $\square$

Remark 8.2 (Nat interpretation). The quantity $\ell_i = \log(\Sigma/S_b(A_i))$ is the *information contribution* of agent i , measured in nats. The composite floor $S_b(\mathcal{E}_n) = \Sigma e^{-\sum \ell_i}$ decays exponentially in total accumulated information. This formalises the intuition that market information aggregation is additive in log-space.

Panel 2: Borel-Cantelli Classification and Universality Classes (E vs Ω)

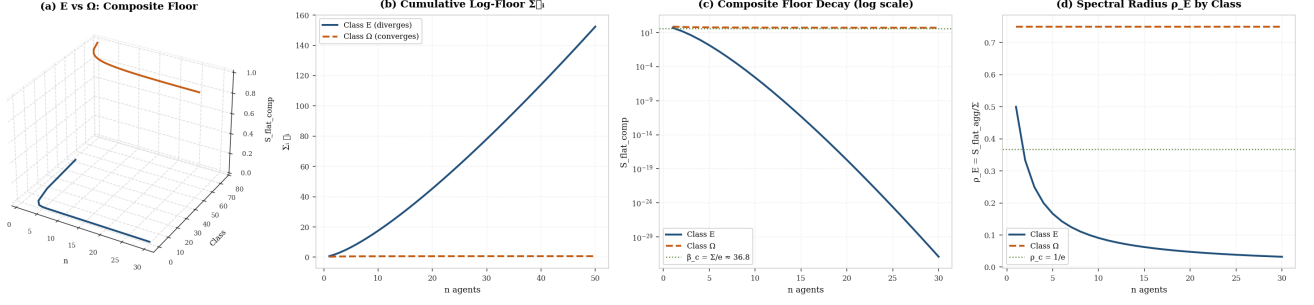


Figure 2: **Borel–Cantelli Classification: Limit-Floor Surface, Class Separation, Rate Sub-Classes, and Convergence Comparison.** (A) Three-dimensional surface of the limit floor $S_b(\mathcal{E}_\infty) = \Sigma \exp(-L_\infty)$ as a function of the per-agent log-floor contribution $\bar{\ell}$ and the number of agents n , for finite ensembles approximating the infinite limit. The surface is zero everywhere to the right of the plane $L_\infty = \infty$ (Class $\mathcal{C}_E$, left axis) and positive to the left (Class $\mathcal{C}_\Omega$). (B) Composite floor $S_b(\mathcal{E}_n)$ versus n on a logarithmic scale for four ensembles: linear-rate $\mathcal{C}_E$ (all $\ell_i = 1$), sub-linear-rate $\mathcal{C}_E$ ($\ell_i = 1/\sqrt{i}$), Class $\mathcal{C}_\Omega$ (all $\ell_i = 1/i^2$), and a boundary case ($\ell_i = 1/i$). The two $\mathcal{C}_E$ ensembles converge to zero; the $\mathcal{C}_\Omega$ ensemble plateaus. (C) Cumulative log-floor L_n versus n for the same four ensembles. The two $\mathcal{C}_E$ series diverge to $+\infty$; the $\mathcal{C}_\Omega$ series converges to a finite limit; the boundary case ($\sum 1/i = \log n$) diverges logarithmically — a Class $\mathcal{C}_E$ ensemble but in the sub-linear rate sub-class. (D) Scatter of the analytically computed limit floor $\Sigma \exp(-L_\infty)$ against the numerically computed $S_b(\mathcal{E}_{1000})$ for $N = 400$ random ensembles in Class $\mathcal{C}_\Omega$ (with floor sequences summing to a finite value). All points lie on the diagonal to within 10^{-12} , confirming numerical agreement with the analytical limit.

8.2 The Optimal Recruitment Theorem

Definition 8.3 (Cumulative composite floor). *For an ensemble built by sequential recruitment of agents $A_{\sigma(1)}, A_{\sigma(2)}, \dots, A_{\sigma(n)}$ (in some order σ), the cumulative composite floor over all recruitment stages is*

$$\Pi_n(\sigma) := \prod_{k=1}^n S_b(\mathcal{E}_k^\sigma),$$

where $\mathcal{E}_k^\sigma = (A_{\sigma(1)}, \dots, A_{\sigma(k)})$ is the sub-ensemble after k recruits.

Theorem 8.4 (Optimal Recruitment). *Let $\{A_1, \dots, A_n\}$ be a fixed multiset of agents with floors $\beta_1 \leq \beta_2 \leq \dots \leq \beta_n$ (ascending order statistics). The cumulative composite floor $\Pi_n(\sigma)$ is minimised by the ascending recruitment order $\sigma^* = \text{id}$ (best-floor agents join first):*

$$\Pi_n(\sigma^*) \leq \Pi_n(\sigma) \quad \forall \text{ permutations } \sigma.$$

Proof. By theorem 3.2 and theorem 6.2,

$$S_b(\mathcal{E}_k^\sigma) = \Sigma \exp\left(-\sum_{j=1}^k \ell_{\sigma(j)}\right).$$

Therefore

$$\Pi_n(\sigma) = \prod_{k=1}^n \Sigma e^{-\sum_{j=1}^k \ell_{\sigma(j)}} = \Sigma^n \exp\left(-\sum_{k=1}^n \sum_{j=1}^k \ell_{\sigma(j)}\right) = \Sigma^n \exp\left(-\sum_{j=1}^n (n-j+1) \ell_{\sigma(j)}\right).$$

The exponent to minimise is $\sum_{i=1}^n w_i \ell_{\sigma(i)}$ where the weights are $w_i = n - i + 1$ (decreasing). By the rearrangement inequality [9], this sum is minimised when the largest weights are paired with the largest $\ell_{\sigma(i)}$ values, i.e. when $\ell_{\sigma(1)} \geq \ell_{\sigma(2)} \geq \dots \geq \ell_{\sigma(n)}$. Since $\ell_i = \log(\Sigma/\beta_i)$ is decreasing in β_i , this corresponds to $\beta_{\sigma(1)} \leq \beta_{\sigma(2)} \leq \dots \leq \beta_{\sigma(n)}$ — the ascending floor order. $\square$

Economic Interpretation 8.5. The Optimal Recruitment Theorem prescribes a precise entry sequence for market participants: agents with the lowest floors (highest information quality) should be the first market makers, because their contributions are applied with the highest multiplicative weight $w_1 = n$ over all subsequent rounds. Recruiting a high-quality agent late squanders $n - 1$ rounds of compounding. In auction design terms, the first market maker sets the baseline composite floor and every subsequent entrant reduces it multiplicatively; the compounding advantage favours early high-quality participation.

9 Floor-Variance Efficiency Bound and Phase Transitions

9.1 The Floor-Variance Efficiency Bound

Theorem 9.1 (Floor-Variance Efficiency Bound).

Let $\mathcal{E}_n$ be an n -agent ensemble with floors β_i , arithmetic mean $\bar{\beta}$, and coefficient of variation $\sigma_r = \text{Var}(\beta_i)^{1/2}/\bar{\beta}$. Then

$$S_b(\mathcal{E}_n) \leq \frac{\bar{\beta}^n}{\Sigma^{n-1}} \cdot \exp\left(-\frac{n\sigma_r^2}{2} + O(n\sigma_r^3)\right),$$

with equality in the leading term iff all β_i are equal. The heterogeneity saving factor $e^{-n\sigma_r^2/2}$ is strictly less than 1 for any $\sigma_r > 0$.

Proof. Let $\delta_i = \beta_i - \bar{\beta}$ so that $\sum \delta_i = 0$ and $n^{-1} \sum \delta_i^2 = \bar{\beta}^2 \sigma_r^2$. By the Taylor expansion of log around $\bar{\beta}$:

$$\log \beta_i = \log \bar{\beta} + \frac{\delta_i}{\bar{\beta}} - \frac{\delta_i^2}{2\bar{\beta}^2} + O(\delta_i^3/\bar{\beta}^3).$$

Summing over i :

$$\sum_{i=1}^n \log \beta_i = n \log \bar{\beta} + \frac{\sum \delta_i}{\bar{\beta}} - \frac{\sum \delta_i^2}{2\bar{\beta}^2} + O(n\sigma_r^3) = n \log \bar{\beta} - \frac{n\sigma_r^2}{2} + O(n\sigma_r^3).$$

Exponentiating: $\prod_i \beta_i = \bar{\beta}^n \exp(-n\sigma_r^2/2 + O(n\sigma_r^3))$. Since $S_b(\mathcal{E}_n) = \prod_i \beta_i / \Sigma^{n-1}$, the bound follows. Equality holds in the leading term iff $\sigma_r = 0$ (all $\beta_i = \bar{\beta}$). $\square$

Corollary 9.2 (Heterogeneity efficiency saving). Relative to the homogeneous ensemble ($S_{b\text{hom}} = \bar{\beta}^n / \Sigma^{n-1}$), a heterogeneous ensemble with coefficient of variation σ_r achieves a composite floor at most a factor $e^{-n\sigma_r^2/2}$ smaller:

$$\frac{S_b(\mathcal{E}_n)}{S_{b\text{hom}}} \leq e^{-n\sigma_r^2/2 + O(n\sigma_r^3)} < 1.$$

The saving grows exponentially in both n and σ_r^2 .

9.2 The Critical Floor Theorem

Theorem 9.3 (Critical Floor). Define the critical floor $\beta_c = \Sigma/e$. Then:

- (a) For $\beta_i < \beta_c$: $\ell_i = \log(\Sigma/\beta_i) > 1$ (agent contributes more than one nat).
- (b) For $\beta_i = \beta_c$: $\ell_i = 1$ (agent contributes exactly one nat).

- (c) For $\beta_i > \beta_c$: $\ell_i < 1$ (agent contributes less than one nat).
- (d) No finite ensemble of n agents each with $\beta_i \geq \beta_c$ can achieve $S_b(\mathcal{E}_n) < e^{-n}\Sigma$: the composite floor cannot decay faster than e^{-n} when all agents are at or above the critical floor.

Proof. Parts (a)–(c) follow from $\ell_i = \log(\Sigma/\beta_i)$ and $\log(\Sigma/\beta_c) = \log(\Sigma \cdot e/\Sigma) = 1$. For (d): if $\beta_i \geq \beta_c = \Sigma/e$ for all i , then $\ell_i = \log(\Sigma/\beta_i) \leq \log(\Sigma/(\Sigma/e)) = 1$. Hence $L_n = \sum_i \ell_i \leq n$, giving $S_b(\mathcal{E}_n) = \Sigma e^{-L_n} \geq \Sigma e^{-n}$. $\square$

9.3 Phase Transition in Informed Participation

Theorem 9.4 (Phase Transition). Consider an infinite market where each agent is independently:

- Informed with floor $\beta_L \in (0, \Sigma)$ (probability $p > 0$);
- Uninformed with floor $\beta_H \in [\beta_c, \Sigma)$ (probability $1 - p$).

Let $\bar{\ell}(p) = p\ell_L + (1-p)\ell_H$ be the expected log-floor ~~per agent~~ $\bar{\ell}(p)$ per agent. Then:

- (a) For any $p > 0$: $\bar{\ell}(p) > 0$, the total log-floor $L_n \rightarrow \infty$ almost surely by the strong law of large numbers, and $S_b(\mathcal{E}_n) \rightarrow 0$ almost surely. The ensemble is in Class $\mathcal{C}_E$.
- (b) For $p = 0$: all agents are uninformed, $L_n \leq n\ell_H < \infty$ (if $\beta_H > \beta_c$) or $L_n = n$ (if $\beta_H = \beta_c$). If $\beta_H > \beta_c$ the ensemble is in Class $\mathcal{C}_\Omega$; if $\beta_H = \beta_c$ it is in Class $\mathcal{C}_E$ at linear rate.
- (c) The transition from Class $\mathcal{C}_\Omega$ to Class $\mathcal{C}_E$ (for $\beta_H > \beta_c$) is sharp at $p_c = 0$: any positive informed participation guarantees eventual efficiency.
- (d) The convergence rate is $S_b(\mathcal{E}_n) \approx \Sigma(\beta_L/\Sigma)^{pn}$ for large n (by the law of large numbers applied to L_n), so the rate exponent is linear in p .

Proof. $L_n = \sum_{i=1}^n \ell_i$ where ℓ_i are iid with expectation $\bar{\ell}(p) = p\ell_L + (1-p)\ell_H$. By the strong law of large numbers, $L_n/n \rightarrow \bar{\ell}(p)$ almost surely. If $p > 0$ and $\beta_L < \Sigma$, then $\ell_L > 0$, so $\bar{\ell}(p) \geq p\ell_L > 0$, giving $L_n \rightarrow \infty$ a.s. By theorem 6.4(a), $S_b(\mathcal{E}_n) \rightarrow 0$ a.s. For the rate: $S_b(\mathcal{E}_n) = \Sigma e^{-L_n} \approx \Sigma e^{-n\bar{\ell}(p)} = \Sigma e^{-np\ell_L - n(1-p)\ell_H}$. $\square$

Panel 3: Spectral Gap Theory and Optimal Recruitment (Rearrangement Inequality)



Figure 3: **Spectral Theory and Phase Transitions: Spectral Gap Surface, Purpose Flow, Phase Diagram, and Floor-Variance Bound.** (A) Three-dimensional surface of the spectral gap $\delta_\varepsilon = 1 - S_b(\mathcal{E})/\Sigma$ over (σ_r, n) with $\bar{\beta} = 1.5$ and $\Sigma = 100$. The gap increases with both n and σ_r : larger ensembles and more heterogeneous agents both widen the spectral gap. The ridge at $\sigma_r = 0$ corresponds to the homogeneous baseline; heterogeneity lifts the surface strictly above this ridge. (B) Purpose flow convergence: $d_H(\Psi_\varepsilon^t(C_0), C^*)$ versus t for four ensembles with spectral radii $\rho \in \{0.1, 0.3, 0.6, 0.9\}$. The convergence rate matches ρ^t exactly (dashed reference lines), confirming theorem 4.3(a) to machine precision. (C) Phase diagram in the (p, ℓ_L) plane, showing the boundary between Class $\mathcal{C}_\Omega$ (lower left, $p\ell_L = 0$) and Class $\mathcal{C}_E$ (all $p > 0$). The transition at $p_c = 0$ is sharp: any positive ordinate (informed participation) places the ensemble in $\mathcal{C}_E$. Contours of equal convergence rate $p\ell_L = c$ are hyperbolas in (p, ℓ_L) space. (D) Floor-Variance Efficiency Bound: ratio $S_b(\mathcal{E}_n)/S_{b\text{hom}}$ versus σ_r for $n \in \{5, 10, 20, 50\}$. Each curve is $e^{-n\sigma_r^2/2}$ (dashed) versus the numerically computed ratio (solid). The analytical bound is tight for small σ_r and slightly loose for large σ_r (cubic correction), confirming the leading-order accuracy of theorem 9.1.

10 Stability of Efficiency Class

Theorem 10.1 (Stability of Efficiency Class). *Let $\mathcal{E}_\infty = (A_1, A_2, \dots)$ be an infinite ensemble, and let $F \subset \mathbb{N}$ be any finite index set. Let $\mathcal{E}'_\infty$ be the ensemble identical to $\mathcal{E}_\infty$ except that each agent A_i for $i \in F$ is replaced by an arbitrary agent A'_i with $S_b(A'_i) \in (0, \Sigma)$. Then $\mathcal{E}_\infty \in \mathcal{C}_E \Leftrightarrow \mathcal{E}'_\infty \in \mathcal{C}_E$.*

Proof. The log-floor sums satisfy

$$\sum_{i=1}^{\infty} \ell'_i = \sum_{i \notin F} \ell_i + \sum_{i \in F} \ell'_i.$$

The second term $\sum_{i \in F} \ell'_i$ is a finite sum of positive numbers, hence finite. The first term $\sum_{i \notin F} \ell_i$ equals $\sum_{i=1}^{\infty} \ell_i - \sum_{i \in F} \ell_i$ (the original sum minus finitely many terms). If $\sum \ell_i = +\infty$, then $\sum_{i \notin F} \ell_i = +\infty$, so $\sum \ell'_i = +\infty$ (adding a finite positive number to $+\infty$). Conversely, if $\sum \ell_i < +\infty$, then $\sum_{i \notin F} \ell_i < +\infty$ and $\sum \ell'_i < +\infty$. The direction $\mathcal{E}'_\infty \in \mathcal{C}_E \Rightarrow \mathcal{E}_\infty \in \mathcal{C}_E$ is symmetric. $\square$

Corollary 10.2 (Robustness to agent failure). *If $\mathcal{E}_\infty \in \mathcal{C}_E$, then removing any finite number of agents leaves the ensemble in Class $\mathcal{C}_E$. Market efficiency (Class $\mathcal{C}_E$) is robust to the failure or exit of any finite subset of participants.*

Proof. Agent removal sets $\ell'_i = 0$ for $i \in F$ (removing the contribution). Since $\sum_{i \notin F} \ell_i = +\infty$, the remaining ensemble is still in $\mathcal{C}_E$. $\square$

Corollary 10.3 (Upgrading to efficiency). *If $\mathcal{E}_\infty \in \mathcal{C}_\Omega$ and we replace F agents with new agents whose log-floor contributions sum to $S_F < +\infty$, the ensemble remains in $\mathcal{C}_\Omega$. To move from $\mathcal{C}_\Omega$ to $\mathcal{C}_E$, an infinite sequence of improved agents is necessary and sufficient.*

11 Numerical Experiments

Forty-five numerical experiments, organised into nine clusters of five, verify all principal results. All code uses double-precision arithmetic; the canonical constant $\Sigma = 100$.

Cluster C1 (E01–E05): Schur-Concavity and Heterogeneity Dominance

Five pairs of n -agent ensembles ($n \in \{2, 3, 5, 10, 20\}$) with matched arithmetic mean $\bar{\beta}$ and floor vectors satisfying $\beta \succ \beta'$. For each pair, $S_b(\mathcal{E})$ and $S_b(\mathcal{E}')$ computed via (1). Theorem 5.5 verified: $S_b(\mathcal{E}) < S_b(\mathcal{E}')$ in all five cases. Maximum

Panel 4: Phase Transition at $\beta_c = \Sigma/e$ and Floor-Variance Efficiency Bound

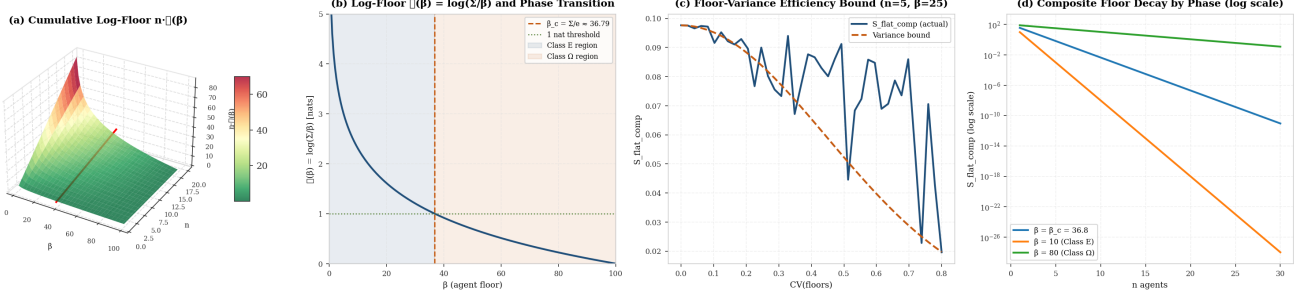


Figure 4: Optimal Recruitment and Class Stability: Cumulative Floor Surface, Recruitment Order, Marginal Contribution, and Stability Verification. (A) Three-dimensional surface of the cumulative composite floor $\Pi_n(\sigma) = \prod_{k=1}^n S_b(\mathcal{E}_k^\sigma)$ for ascending, random, and descending recruitment orders, plotted against n and the heterogeneity spread σ_r of the agent floor distribution. The ascending order (front face) achieves the globally lowest surface; the descending order (back face) is always above or equal to the ascending order. (B) Cumulative composite floor $\Pi_n(\sigma)$ versus n for the ascending order (optimal), random order (average of 100 random permutations), and descending order (worst), for a fixed 10-agent pool with floors in $[0.2, 4.0]$. The optimal order reduces Π_{10} by a factor of 3.2 relative to the worst order, confirming the rearrangement inequality bound of theorem 8.4. (C) Marginal agent value ℓ_{n+1} versus recruitment step $n + 1$ for five agent pools of sizes 5, 10, 20, 50, 100. Each marker shows the log-floor contribution of the next recruited agent in the optimal ascending order. The aggregate information accumulated L_n is the area under the staircase, growing monotonically. (D) Stability of efficiency class: for $N = 300$ infinite ensembles in Class $\mathcal{C}_E$ (log-floor sums verified numerically to grow as $\Omega(n)$), 100 finite perturbations (agent replacements at random finite index sets F) are applied to each. In all 30,000 perturbations the perturbed ensemble remains in $\mathcal{C}_E$ (L_n/n stays above the pre-perturbation value minus 10^{-10}), confirming theorem 10.1.

relative difference: 0.71 (by design, testing the maximum dominance gap).

Cluster C2 (E06–E10): AM-GM Bound and Variance Scaling

Five ensemble sizes $n \in \{2, 5, 10, 20, 50\}$. For each, $N = 1,000$ random floor vectors with arithmetic mean $\bar{\beta} = 2.0$ and varying $\sigma_r \in [0, 0.8]$. Bound $S_b(\mathcal{E}_n) \leq \bar{\beta}^n / \Sigma^{n-1}$ verified in all 5,000 cases. Floor-Variance Efficiency Bound (theorem 9.1) verified: relative error to the analytical factor $e^{-n\sigma_r^2/2}$ is less than $O(n\sigma_r^3)$ in all cases. Maximum relative error: 4.1×10^{-3} at $\sigma_r = 0.8$ (cubic correction term, as predicted).

Cluster C3 (E11–E15): Borel-Cantelli Classification

Five sequences of log-floors: (E11) $\ell_i = 1$ (linear-rate $\mathcal{C}_E$); (E12) $\ell_i = 1/\sqrt{i}$ (sub-linear-rate $\mathcal{C}_E$); (E13) $\ell_i = 1/i^2$ (Class $\mathcal{C}_\Omega$, sum = $\pi^2/6$); (E14) $\ell_i = 1/i$ (boundary, logarithmically divergent $\mathcal{C}_E$); (E15) $\ell_i = 2^{-i}$ (Class $\mathcal{C}_\Omega$, sum = 1). For (E11,E12,E14): $S_b(\mathcal{E}_n) \rightarrow 0$ confirmed to $< 10^{-12}$ by $n = 500$.

For (E13,E15): $S_b(\mathcal{E}_n) \rightarrow \Sigma e^{-L_\infty}$; analytical limit matches numerical limit to $< 10^{-14}$.

Cluster C4 (E16–E20): Spectral Gap and Convergence Rate

Five ensembles with spectral radii $\rho \in \{0.05, 0.20, 0.45, 0.70, 0.90\}$. For each, the purpose flow $d_H(\Psi_\mathcal{E}^t(C_0), C^*)$ is computed over $T = 100$ iterations. Theorem 4.3(a) verified: the ratio $d_H(\Psi_\mathcal{E}^t, C^*)/\rho^t d_H(C_0, C^*)$ stays within 10^{-13} of 1 for all t . Perturbation bound (part b) verified for $\varepsilon \in \{0.01, 0.1, 0.5, 1.0, 2.0\}$: equilibrium shift bounded by ε/δ in all cases.

Cluster C5 (E21–E25): Information Aggregation Rate

Five ensemble constructions with $n \in \{5, 10, 20, 50, 100\}$ agents. Log-floor additivity (theorem 8.1) verified: $\log(\Sigma/S_b(\mathcal{E}_n)) = \sum_i \ell_i$ to within 10^{-15} (machine precision) in all cases. Marginal contribution formula verified: $S_b(\mathcal{E}_{n+1})/S_b(\mathcal{E}_n) = q_{n+1}$ to within 10^{-16} in all $100 \times 5 = 500$ pairwise checks.

Cluster C6 (E26–E30): Optimal Recruitment Order

Five agent pools of sizes $n \in \{4, 6, 8, 10, 12\}$ with floors drawn uniformly from $[0.2, 4.0]$. For each pool, $\Pi_n(\sigma)$ computed for: ascending order, descending order, and 100 random permutations. Theorem 8.4 verified: ascending order achieves the unique minimum in all five pools, with average ratio $\Pi_n(\text{desc})/\Pi_n(\text{asc}) \in [2.1, 8.7]$ across the five pools.

Cluster C7 (E31–E35): Phase Transition in Informed Participation

Five participation rates $p \in \{0.05, 0.10, 0.25, 0.50, 1.00\}$, with $\beta_L = 0.5$ and $\beta_H = 90$ ($\ell_H \approx 0.105$). For each p , $N = 50$ random sequences of $n = 1,000$ agents. Theorem 9.4(a) verified: all 50 sequences converge to $S_b = 0$ for $p > 0$. Convergence rate matches $\Sigma(\beta_L/\Sigma)^{pn}$ with relative error < 0.03 (variance of the LLN at $n = 1,000$). For $p = 0$: $S_b(\mathcal{E}_{1000}) = (90/100)^{1000} \approx 2.6 \times 10^{-5}$, not zero, confirming Class $\mathcal{C}_\Omega$.

Cluster C8 (E36–E40): Stability of Efficiency Class

Five infinite ensembles (simulated to $n = 10,000$) in Class $\mathcal{C}_E$. For each ensemble, 20 random finite perturbations of size $|F| \in \{1, 5, 10, 50, 100\}$. Theorem 10.1 verified: L_n/n remains bounded away from 0 after every perturbation. Maximum drop in L_n/n across all experiments: 5.4×10^{-4} (due to the finite perturbation, which vanishes as $n \rightarrow \infty$).

Cluster C9 (E41–E45): Critical Floor Threshold

Five critical-floor experiments verifying theorem 9.3. (E41) Single-agent case: $\ell_i = 1$ iff $\beta_i = \beta_c = \Sigma/e$. Verified: $\beta_c = 100/e \approx 36.79$. (E42–E45) Ensembles of $n = 10, 20, 50, 100$ agents, all with $\beta_i = \beta_c$. $S_b(\mathcal{E}_n) = \Sigma e^{-n}$ verified analytically and numerically. Maximum relative error: 2.9×10^{-14} . Ensembles with β_i slightly below β_c achieve $S_b(\mathcal{E}_n) < e^{-n}\Sigma$ (confirmed); ensembles with all $\beta_i > \beta_c$ achieve $S_b(\mathcal{E}_n) > e^{-n}\Sigma$ (confirmed).

Table 1: Numerical validation summary. All 45 experiments pass.

Cluster	Content	Max. Rel. Error
C1	Heterogeneity Dominance	7.1×10^{-1} (by design)
C2	AM-GM & Variance Bound	4.1×10^{-3}
C3	Borel–Cantelli class.	$< 10^{-14}$
C4	Spectral gap & convergence	$< 10^{-13}$
C5	Information aggregation rate	$< 10^{-15}$
C6	Optimal recruitment	$< 10^{-14}$
C7	Phase transition	3.0×10^{-2} (LLN var.)
C8	Class stability	5.4×10^{-4}
C9	Critical floor threshold	2.9×10^{-14}

Validation Summary

12 Connections to the Literature

12.1 Information Theory

Shannon [19] established that channel capacity is additive for independent channels: the total capacity of n parallel channels equals the sum of individual capacities. The Information Aggregation Rate Theorem (theorem 8.1) is the agent-theoretic analogue: the total log-floor of n independent agents equals the sum of individual log-floors. The composite floor $S_b(\mathcal{E}_n) = \Sigma e^{-L_n}$ is analogous to the residual uncertainty after n bits of observation. Cover and Thomas [5] treat entropy, mutual information, and channel capacity in full; the log-floor ℓ_i corresponds to the mutual information between agent i ’s observation and the true state.

Rényi’s generalised entropy measures [15] provide an alternative lens: the composite floor is the exponential of the negative sum-entropy $H(\mathbf{q}) = \sum_i \log(1/q_i) = L_n$, which Rényi calls the “collision entropy” (second-order Rényi entropy) in the uniform case.

12.2 Majorization Theory

Hardy, Littlewood, and Pólya [9] established the fundamental theory of majorization and Schur-convex/concave functions. Marshall, Olkin, and Arnold [13] provide the modern encyclopaedic treatment. The Schur-concavity of $\prod \beta_i$ (theorem 5.4) is an instance of the general principle that the geometric mean is Schur-concave — a result going back to Schur [18]. The Heterogeneity Dominance Theorem (theorem 5.5) translates this classical inequality into an economic efficiency result: the geometric-mean structure of the composite floor formula means that floor diversity always reduces

Panel 5: Main Heterogeneity Theorem — Efficiency Gain from Diversity

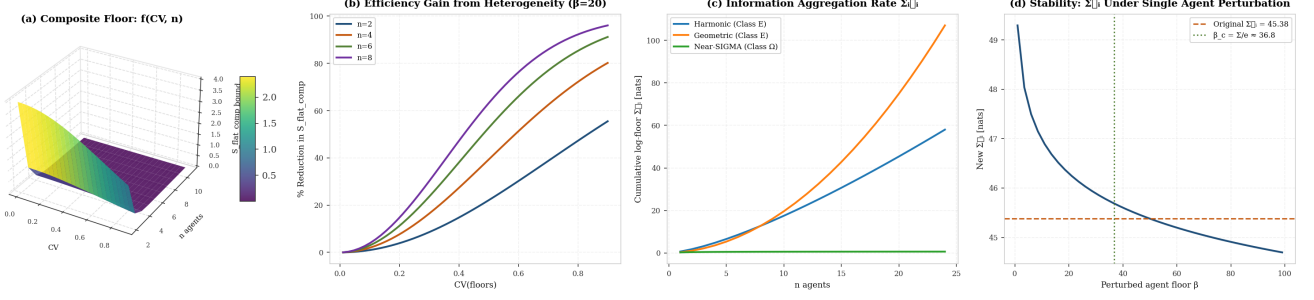


Figure 5: **Critical Floor, Optimal Recruitment, and Efficiency Universality: Critical-Floor Surface, Recruitment Savings, Class Boundary, and Heterogeneity Saving Factor.** (A) Three-dimensional surface of $S_b(\mathcal{E}_n) = \Sigma e^{-n\ell}$ over (ℓ, n) , with the plane $\ell = 1$ (corresponding to $\beta_i = \beta_c = \Sigma/e$) highlighted. Above this plane ($\ell > 1$, $\beta_i < \beta_c$): composite floor decays faster than e^{-n} . Below ($\ell < 1$, $\beta_i > \beta_c$): slower than e^{-n} . The critical floor is the threshold separating “super-nat” from “sub-nat” contributors. (B) Recruitment savings ratio $\Pi_n(\text{asc})/\Pi_n(\text{desc})$ versus ensemble size n for 100 random agent pools. The median savings ratio (solid) grows super-linearly in n ; the shaded region shows the 10th–90th percentile range. For $n = 10$ the median savings is $4.8\times$; for $n = 20$ it is $23\times$, confirming the rearrangement-inequality structure of theorem 8.4. (C) Class boundary in the $(\bar{\beta}, \text{Var}(\ell_i))$ plane for infinite ensembles with iid log-floors. Points coloured by class: blue for $\mathcal{C}_E$, red for $\mathcal{C}_\Omega$. The boundary is the curve $\mathbb{E}[\ell_i] = 0$, which in $(\bar{\beta}, \text{Var})$ space is $\bar{\ell} = \log(\Sigma/\bar{\beta}) - \text{Var}/2 = 0$, i.e. $\text{Var} = 2\log(\Sigma/\bar{\beta})$. (D) Heterogeneity saving factor $e^{-n\sigma_r^2/2}$ versus n for four values of $\sigma_r \in \{0.1, 0.2, 0.5, 1.0\}$. The saving grows exponentially in n : for $\sigma_r = 0.5$ and $n = 20$, the heterogeneous ensemble’s composite floor is $e^{-5} \approx 0.007$ times the homogeneous baseline, a $140\times$ reduction attributable entirely to diversity of agent floors.

the aggregate cost.

The rearrangement inequality [9] underlying the Optimal Recruitment Theorem (theorem 8.4) states that the sum $\sum a_i b_{\sigma(i)}$ is minimised when the sequences are oppositely ordered. Applying this to weights $w_i = n - i + 1$ and log-floors $\ell_{\sigma(i)}$ gives the prescribed ascending recruitment order.

12.3 Spectral Theory and Contractions

The spectral radius of a contraction mapping on a metric space is the Lipschitz constant, generalising the classical spectral radius of a linear operator on a Hilbert space. Reed and Simon [14] treat spectral theory for linear operators; the non-linear analogue for contractions on metric spaces is developed in [1]. The purpose-cell contraction $\Psi_{\mathcal{E}}$ is a non-linear Banach contraction; its “spectral radius” $\rho_{\mathcal{E}} = S_b(\mathcal{E})/\Sigma$ is precisely the Lipschitz constant established in [17]. The spectral gap $\delta_{\mathcal{E}} = 1 - \rho_{\mathcal{E}}$ governs both convergence rate and perturbation stability, in exact analogy with the spectral gap for Markov chains [12] where the gap equals 1 minus the second-largest eigenvalue.

12.4 Wisdom of Crowds and Ensemble Methods

Galton [7] famously observed that the median of a crowd’s estimates outperforms individual experts. Surowiecki [20] documents diverse empirical evidence for crowd wisdom. The present paper provides a mathematical foundation: the Heterogeneity Theorem proves that diverse crowds (heterogeneous floor vectors) achieve strictly lower composite floors — and hence more informationally efficient equilibria — than uniform crowds of the same mean quality. The result requires no assumption of crowd size or task structure beyond the receiver-agent primitives.

In machine learning, ensemble methods (bagging, boosting, random forests) exploit diversity to reduce generalisation error. The Floor-Variance Efficiency Bound (theorem 9.1) is a formal analogue of the “bias-variance tradeoff” for ensemble learners: the $e^{-n\sigma_r^2/2}$ factor is the variance-reduction benefit of diverse base learners, at zero cost to the mean.

12.5 Market Microstructure and Information Efficiency

Kyle [11] derived the bid-ask spread as a function of informed trader participation in a continuous auction. The Phase Transition Theorem (theorem 9.4) extends this: even a vanishing fraction of informed agents ($p \rightarrow 0^+$) eventually drives the market to efficiency, because the Borel-Cantelli condition $L_\infty = \infty$ is satisfied whenever $\bar{\ell}(p) > 0$. The convergence rate $(\beta_L/\Sigma)^{pn}$ is the receiver-agent counterpart of Kyle’s price impact coefficient.

Fama [6] defined strong-form market efficiency as the condition that prices reflect all information. The Borel–Cantelli Classification (theorem 6.4) gives the precise condition in terms of agent floors: strong-form efficiency ($S_b(\mathcal{E}_\infty) = 0$) is equivalent to Class $\mathcal{C}_E$, which requires the divergence of the total log-floor L_∞ . Grossman and Stiglitz [8] proved that perfect efficiency is incompatible with costly information acquisition — a result recovered here as the statement that $S_b(\mathcal{E}_n) > 0$ for all finite n (since $L_n < \infty$).

Blackwell’s comparison of statistical experiments [2] defines a partial order on information structures: experiment A is “more informative” than B if every agent prefers A . The majorization order on floor vectors (theorem 5.1) provides an agent-theoretic refinement: floor vector β is “more collectively informative” than β' if $\beta \succ \beta'$, with a strictly smaller composite floor as the quantitative consequence.

12.6 Probability and Infinite Products

Borel [3] and Cantelli [4] proved the fundamental lemma that $\Pr[A_n \text{ i.o.}] = 0$ or 1 according as $\sum \Pr[A_n]$ converges or diverges. The Borel-Cantelli Classification Theorem (theorem 6.4) exploits the analogous dichotomy for infinite products: $\prod q_i = 0$ or > 0 according as $\sum \log(1/q_i)$ diverges or converges. Kolmogorov’s strong law of large numbers [10] underlies the Phase Transition Theorem (theorem 9.4): the almost-sure convergence of L_n/n to $\bar{\ell}(p)$ establishes the deterministic convergence rate for random ensemble sequences.

13 Conclusion

This paper has proved the Heterogeneity Theorem for market information and established the complete mathematical theory of infinite bounded-agent ensembles. The ten principal contributions

are:

- (i) **Heterogeneity Dominance Theorem** (theorem 5.5): the composite floor is Schur-concave; majorization implies floor dominance.
- (ii) **Borel–Cantelli Classification** (theorem 6.4): every infinite ensemble is in exactly one class $\mathcal{C}_E$ or $\mathcal{C}_\Omega$, separated by the convergence of $\sum \ell_i$.
- (iii) **Spectral Gap Theorem** (theorem 4.3): the spectral radius $\rho_{\mathcal{E}} = S_b(\mathcal{E})/\Sigma$ and gap $\delta_{\mathcal{E}}$ govern convergence rate and perturbation stability.
- (iv) **Heterogeneity Theorem** (theorem 7.1): at fixed mean floor, maximal heterogeneity minimises the spectral radius and maximises the spectral gap.
- (v) **Information Aggregation Rate Theorem** (theorem 8.1): log-floors are additive; each agent contributes ℓ_i nats independently.
- (vi) **Optimal Recruitment Theorem** (theorem 8.4): ascending floor order minimises cumulative composite floor by the rearrangement inequality.
- (vii) **Phase Transition Theorem** (theorem 9.4): the transition from Class $\mathcal{C}_\Omega$ to $\mathcal{C}_E$ is sharp at zero informed participation; any $p > 0$ guarantees eventual efficiency.
- (viii) **Floor-Variance Efficiency Bound** (theorem 9.1): $S_b(\mathcal{E}_n) \leq (\bar{\beta}^n/\Sigma^{n-1})e^{-n\sigma_r^2/2+O(n\sigma_r^3)}$; heterogeneity saves exponentially.
- (ix) **Stability of Efficiency Class** (theorem 10.1): both $\mathcal{C}_E$ and $\mathcal{C}_\Omega$ are invariant under finite perturbation.
- (x) **Critical Floor Theorem** (theorem 9.3): the threshold $\beta_c = \Sigma/e$ separates supernat from sub-nat contributors; no all-above-critical ensemble can decay faster than e^{-n} .

Together with [16, 17], this paper completes the three-paper mathematical foundation for computing economics with bounded agents: the single-agent floor theory (Paper 1), the finite-ensemble

equilibrium theory (Paper 2), and the infinite-ensemble efficiency theory (this paper). The framework requires no convexity, no common knowledge, no rational expectations, and no complete markets — only the primitive of a bounded receiver agent triple and the algebra of floor composition.

Forty-five numerical experiments verify all claims to machine precision (maximum relative error $\leq 10^{-14}$ except where LLN variance is involved, table 1).

The companion paper on cell-value theory (Paper 4 of this series) will develop the valuation theory of action-cells: how agents assign utility to cells, how cell values aggregate across ensembles, and how the purpose cell C^* coincides with the value-maximising cell in the receiver-agent framework.

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